

those in need, as discussed earlier (i.e. presumptive eligibility and retroactive coverage). As further evidence of this, a recent study by [Garthwaite et al. \(2019\)](#) finds that individuals moving from uninsured to Medicaid status in ACA Medicaid expansion states had higher than average hospital and ED visits during the pre-ACA period.

We estimate an upper bound for this counterfactual rate for the compliers using observed mortality for actual enrollees in our sample. We first estimate average annual mortality for enrollees in the expansion states during the post-period at 2.0 percent. We consider this an upper bound because the CMS enrollment data does not identify the eligibility pathway for coverage and some of these individuals likely would have qualified for coverage even if there were no eligibility expansions (e.g. if they became disabled);³⁵ we therefore expect that this group likely exhibits worse health than those who are induced to enroll by the policy change.

We then take into account the mortality reduction resulting from the gain in Medicaid coverage. Our analysis suggests that a year of Medicaid coverage is associated with a reduction in annual mortality of 0.35 percentage points (as discussed above). On average, those who were not enrolled in Medicaid prior to the ACA, but did enroll in expansion states after the ACA, remain enrolled for about 2.7 years over our sample period. If compliers experienced 2.7 additional years of coverage due to the expansions, their counterfactual mortality rate would be 2.95 percent ($=2.0 + 2.7 \times 0.35$).

Given these calculations, we may therefore expect the counterfactual mortality rate for the compliers to fall somewhere above the estimate for all sample members of 1.63 percent, but below this upper bound of 2.95 percent. Combined with our estimated treatment effect of a 0.35 percentage point reduction in mortality associated with 1 year of accumulated Medicaid enrollment, these counterfactual rates suggest that one year of Medicaid coverage reduces individual mortality by between 11.9 and 21.5 percent. Meanwhile, our estimate of the treatment effect for contemporaneous Medicaid enrollment (1.03pp) indicates a resulting reduction in individual mortality of between 34.9 and 63.2 percent.

If we expect that the mortality effects of Medicaid are solely driven by changes in access to care for individuals who would otherwise be uninsured, we may prefer to scale our reduced form estimates by the change in net insurance coverage, rather than the change in Medicaid coverage. This calculation implicitly assumes that there is no difference between private insurance coverage and Medicaid that would meaningfully impact mortality risk. If coverage only affects mortality during the year in which the respondent is enrolled (i.e., no longer term effects), the change in contemporaneous insurance coverage (a 4.4 percentage point increase in overall insurance, see [Table I](#)) implies much larger treatment effects, ranging from 102 to 184 percent reductions in individual mortality.

These large estimate sizes suggest that this scaling may miss important effects for those who enrolled in Medicaid as a result of the expansions but would have otherwise enrolled in exchange or employer-based coverage. For example, there may be significant financial differences in the cost of coverage or medical care that affect health care utilization decisions and household finances, both of which may affect mortality risk.³⁶ There may also be differences in the types of services covered.

³⁵Note that those who were disabled prior to the ACA should be excluded as our sample criteria excludes those reporting SSI receipt.

³⁶Medicaid is typically free for beneficiaries with no or minimal cost-sharing for service receipt. In contrast, private insurance coverage typically requires an annual premium, as well as additional cost-sharing. A recent study estimates that, under the ACA Medicaid expansions, new Medicaid enrollees ages 50 to 64 saved approximately \$3,100 in out-of-pocket expenditures each year ([McInerney et al., 2020](#)).

Unlike most private insurers, Medicaid covers many long-term home- and community-based services and supports (Rudowitz et al., 2019), which may be important for individuals with complex health needs and high mortality risk. Alternatively, this calculation may miss other spillover effects among Medicaid non-recipients, or the presence of cumulative effects of insurance coverage that are not captured in contemporaneous coverage measures.³⁷ Finally, it is important to note that the confidence interval on the estimated mortality effect includes much smaller reductions in mortality.

VIII.B Comparisons to Previous Estimates

In this section, we compare the treatment effects that we estimate to those published in the literature. In order to take into account differences in coverage changes and population mortality rates across the different study settings, we compare the proportional average treatment effects for new Medicaid enrollees or for the newly insured.

First, we use the public-use replication kit for the OHIE to examine the effect of Medicaid coverage on participants who were ages 55-64 at the time of the experiment to derive estimates comparable to those presented here (reported in Appendix Table A12). Among this age group, ever receiving Medicaid reduced the probability of mortality over a 16 month period by 1.65 percentage points, or a decline of 71.7 percent relative to the control mean, although this estimate is not statistically significant (associated with a p-value of 0.128).³⁸ This estimated proportional treatment effect is very similar to the estimate in our analysis (i.e. 63 percent) described in the last section.

We also use OHIE data to estimate the effect of insurance coverage, rather than Medicaid coverage, on mortality. We do this by examining the subsample of participants who responded to the OHIE survey and provided information on their insurance status at an initial survey (completed an average of 1 month after coverage approval), or at a survey approximately one year later (an average of 13 months after coverage approval, see Finkelstein et al., 2012). Using the gain in coverage measured in the initial survey, we find a decrease in mortality of 326 percent ($p=0.071$) when compared to the control group mean. Using the 12-month measure, we find a 335 percent mortality reduction ($p=0.153$).³⁹ These are both larger than our estimate of the impact of new insurance coverage under the ACA Medicaid expansions, although the confidence intervals overlap with our estimate. See Appendix Section 7 for further discussion.

Recent work by Goldin et al. (2021) also provides experimental evidence on the impact of health insurance on mortality. Taking advantage of changes in insurance status generated by randomly assigned informational letters, the authors find that each additional month of health insurance coverage reduced mortality in their sample of 45 to 64 year olds by 0.178 percentage points over a two-year period, or 10.1 percent compared to their estimated complier baseline mortality rate. Assuming the effect of coverage decays geometrically, the authors estimate that 12 months of coverage results in a reduction in mortality over the two-year period of 106 percent relative to their estimated complier

³⁷When we consider cumulative exposure to insurance coverage using estimates from our HRS sample, the implied treatment effects indicate that each year enrolled in any coverage reduces mortality by between 12 and 21 percent. However, these estimates are fairly imprecise.

³⁸We can also scale this effect by 12/16th to arrive an annual effect of Medicaid on mortality of 1.24 percentage points. This is comparable, but larger, than the 1.03 percentage point treatment effect of contemporaneous Medicaid enrollment estimated in our analysis.

³⁹Note that baseline mortality for the control group is lower among survey respondents since participants had to be alive in order to respond to the survey.

mean, which is in the range of our estimates for the effects of contemporaneous insurance coverage.

In addition to these two experiments, there are also several quasi-experimental analyses that examine the effects of insurance coverage expansions on non-elderly adult mortality. To facilitate comparisons across these studies, we estimate the implied mortality effects for individuals either gaining Medicaid or insurance coverage as proportionate changes over a counterfactual mortality rate. Studies often vary in how they treat measurement error in survey reports of Medicaid enrollment and in the assumed counterfactual mortality rate; we therefore apply a consistent set of adjustments across these studies. We also present 95 percent confidence intervals for these estimates. Further details on the approach used to construct these scaled estimates and their confidence intervals may be found in Appendix Section 7.

The resulting estimates are reported in Appendix Table A13. It is clear from this table that our estimates are in the ballpark of those from existing work examining the mortality effects of Medicaid or insurance expansions for low-income adults. [Sommers \(2017\)](#) examines changes in mortality among 20 to 64 year-olds following pre-ACA Medicaid expansions in Arizona, Maine, and New York. Using a difference-in-differences design, he finds a mortality reduction of 64.5 percent among individuals gaining insurance coverage. Similarly, taking advantage of county-level variation in the impact of the Massachusetts health reform of 2006, [Sommers et al. \(2014\)](#) find that insurance coverage reduces mortality by 27.3 percent among 20 to 64 year olds. Using mortality records aggregated to the county level, [Borgschulte and Vogler \(2020\)](#) find adults age 20 to 64 experienced a 47.3 percent reduction in mortality under the ACA Medicaid expansions. In our analysis, we find that new insurance coverage reduces mortality by 50 percent among those in our sample age 19 to 64 (although this effect was not statistically significant), in line with the estimates from these other studies. [Chen \(2019\)](#), however, finds a much smaller and statistically insignificant effect on the order of 9 percent for adults ages 25-64 under his analysis of the ACA Medicaid expansions.

Fewer papers include estimates for the age group that we study.⁴⁰ [Black et al. \(2019\)](#) and [Chen \(2019\)](#) find smaller implied mortality effects of individual insurance coverage when compared to our study. The estimate in [Black et al. \(2019\)](#), however, is imprecisely estimated with a large confidence interval inclusive of our estimate.

To summarize, the existing experimental literature on the effect of health insurance on mortality shows large effects, suggesting that reductions in mortality associated with insurance well exceed the average mortality rate of the control group. Such large effects may be due to adverse selection, i.e. that those who sign up for health insurance are particularly vulnerable and would have experienced high mortality rates in the absence of insurance. Evidence from the quasi-experimental literature tends to find smaller, but still substantial (9-65 percent), reductions in mortality associated with coverage. Our estimates tend to be smaller than those documented in the experimental literature but somewhat larger than those reported in quasi-experimental studies.

IX Conclusion

Low-income adults in the United States experience dramatically worse health and reduced longevity as compared to their higher income counterparts. The ACA expansion of Medicaid eligibility to poor

⁴⁰While some studies present subgroup analyses by age, they do not include the corresponding first stage estimates needed to estimate the implied individual treatment effect.

and near-poor adults represents a historic effort to improve access to high-quality medical care for this population. Robust evidence that Medicaid increases the use of effective medical care indicates that the ACA expansion could play a crucial role in reducing income disparities in health among U.S. adults. However, the magnitude of any health improvements experienced by beneficiaries remains largely uncertain, in part due to the limited amount of data available that contains information on both socioeconomic status and objective measures of health.

In this paper, we evaluate these expansions by leveraging data linkages across large-scale federal survey and administrative data. These data linkages allow us to overcome several empirical challenges that previous research in this area encountered, including the lack of information about individual-level characteristics that determine Medicaid eligibility and the systematic mis-reporting of Medicaid enrollment that occurs in survey data. Using these data, we show that the ACA Medicaid expansions substantially reduced mortality rates among those who stood to benefit the most.

Our estimated change in mortality for our sample translates into sizeable gains in terms of the number of lives saved under Medicaid expansion. Using the ACS, we calculate that there are about 3.7 million individuals who meet our sample criteria living in expansion states. Our results therefore indicate that approximately 4,800 fewer deaths occurred per year among this population due to Medicaid expansion, or roughly 19,200 fewer deaths over the first four years alone. This calculation relies on the assumption used throughout the paper; namely, that in the absence of the ACA expansions, mortality in expansion and non-expansion states would have trended similarly.

Our estimates also reveal some information about the potential cost of state decisions not to expand Medicaid. These calculations necessarily rely on the assumption that the effects of Medicaid expansion in the non-expansion states would be similar to those observed in the expansion states. The change in mortality in non-expansion states if they were to adopt the expansion could be larger (e.g., if the beneficiaries in the non-expansion states have worse baseline access to effective health care) or smaller (e.g., if the health system has less capacity to treat the newly insured in non-expansion states). Given these caveats, we estimate that, as there are approximately 3 million individuals meeting this sample criteria in non-expansion states, failure to expand in these states resulted in 15,600 additional deaths over this four year period that could have been avoided if the states had elected to expand coverage.